

Hardware Redesign and Performance Evaluation of a Low-Cost Autonomous Line-Following Robot

Katherine Zhang | Supervisor: Professor Hamid Timorabadi
Department of Electrical and Computer Engineering, University of Toronto

Project Overview

Autonomous infrared (IR) line-following robots introduce students to sensing, embedded programming, control systems, and mechanical design. Using robotics as a teaching tool can greatly enhance the learning of STEM-related concepts across all levels of education [1].

This project redesigns an existing educational line follower using lower-cost drivetrain components:

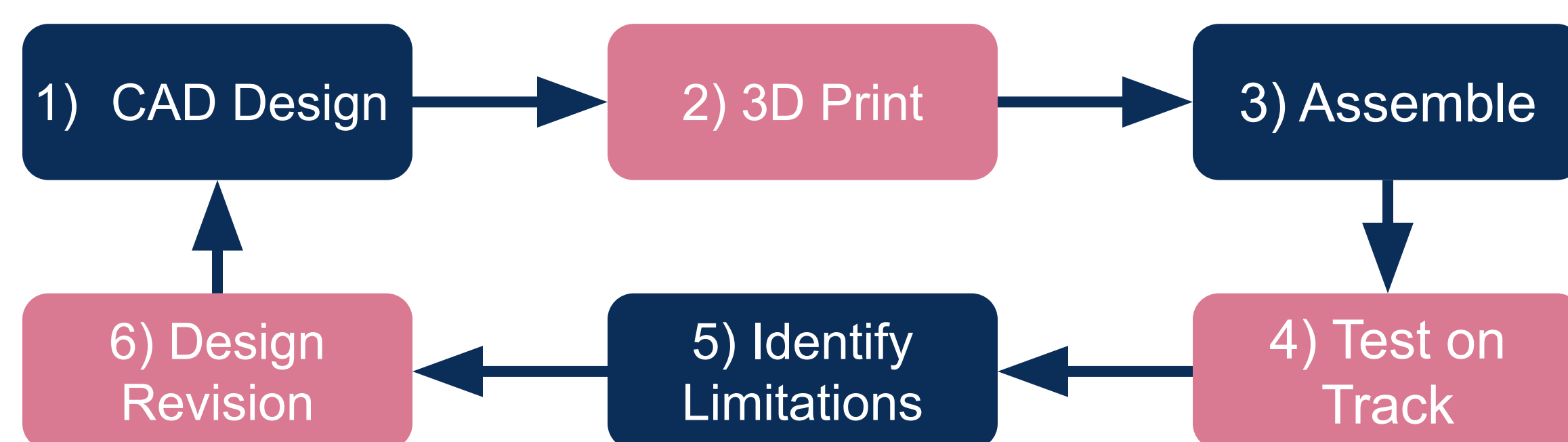
- **Original:** 4 x continuous servos
- **Redesigned:** 2 x DC geared motors + L298N H-bridge + passive caster wheel

The IR sensor array was repositioned to increase look-ahead distance and improve corrective steering. Previous studies have shown that forward track sensing can help anticipate upcoming curves [2].

Research Question

How effective is a redesigned chassis in accommodating lower-cost drivetrain components and reliable autonomous line following for future use in first-year ECE workshops?

Methods

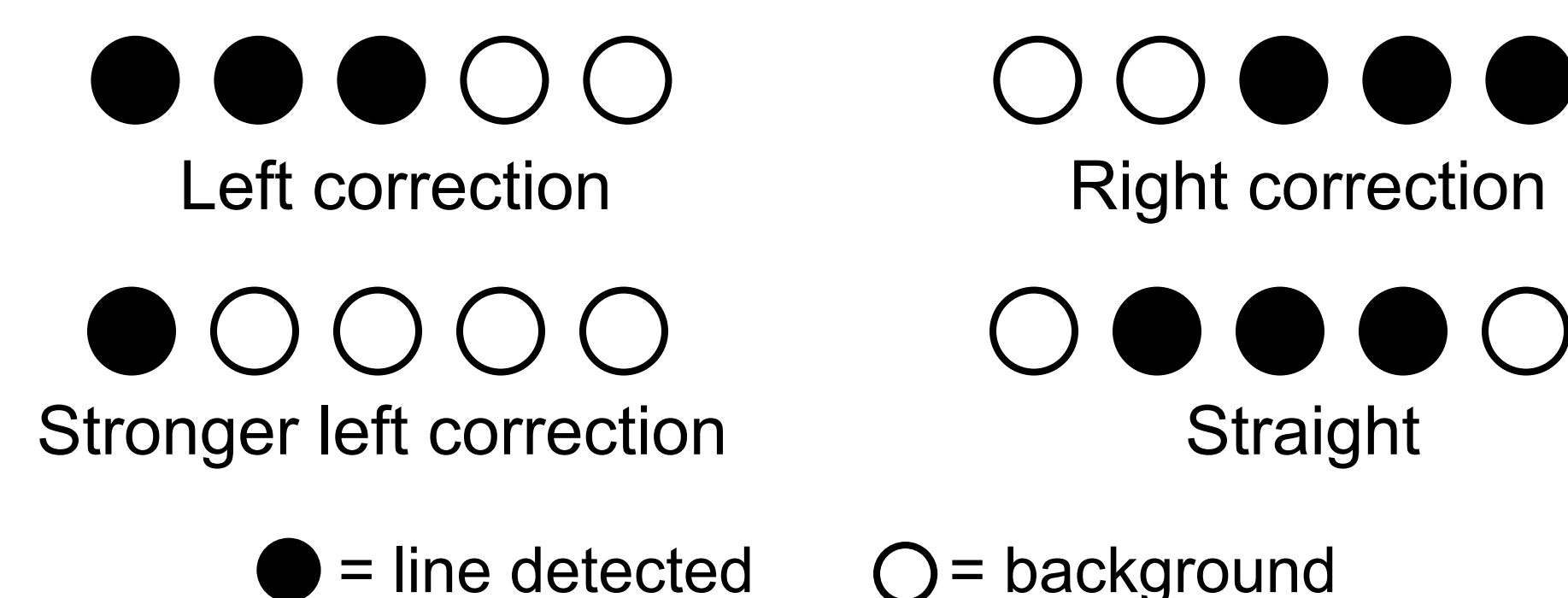


Iterative design process to improve mechanical fit, caster wheel mounting, sensor placement, and autonomous line tracking.

Track testing procedure (after reliable steering logic):

- 25 trials per configuration under consistent conditions
- Trial considered successful if robot completed the full track without losing the line
- Number of successful trials recorded

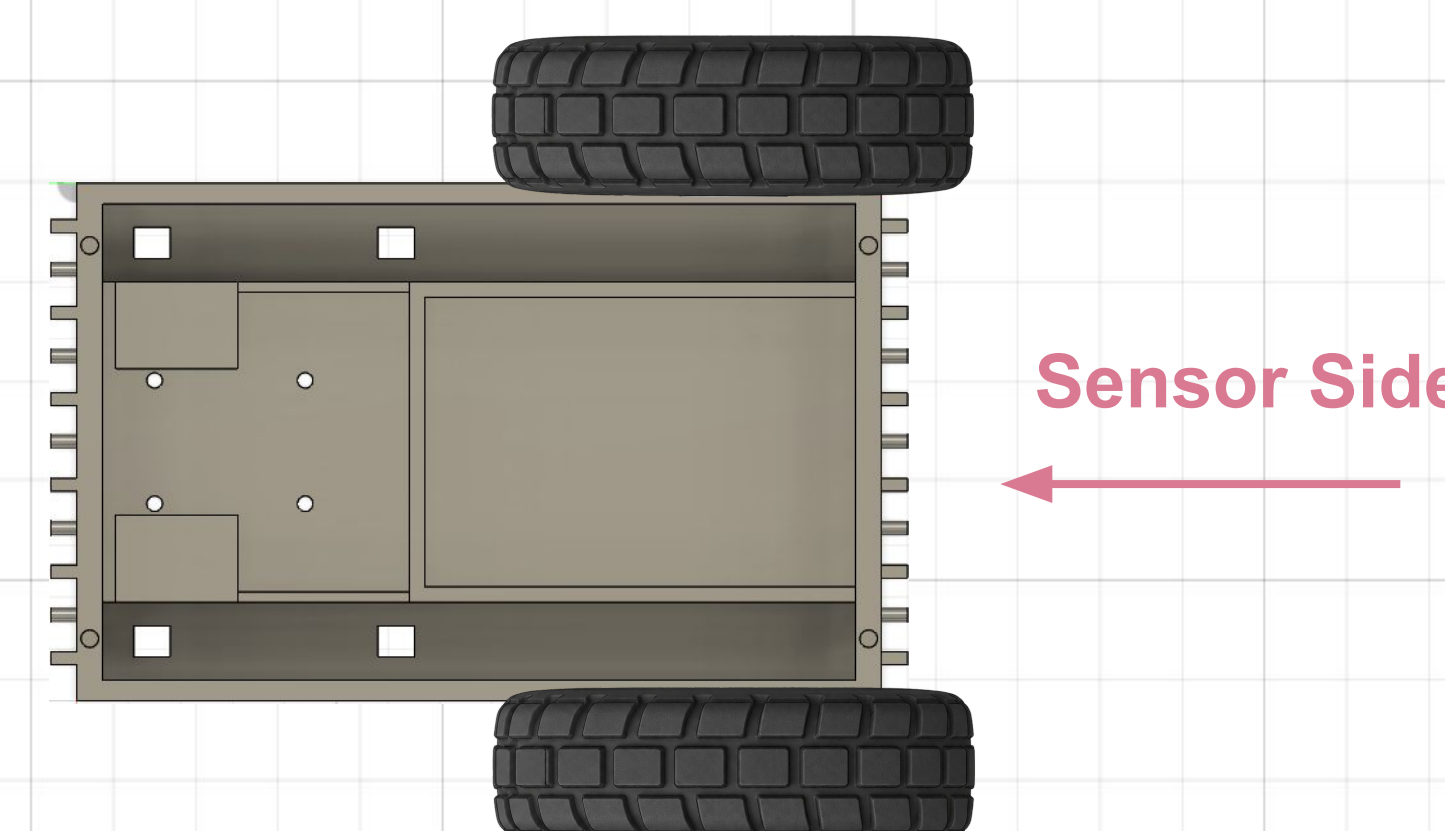
PWM Steering Logic



Results and Discussion

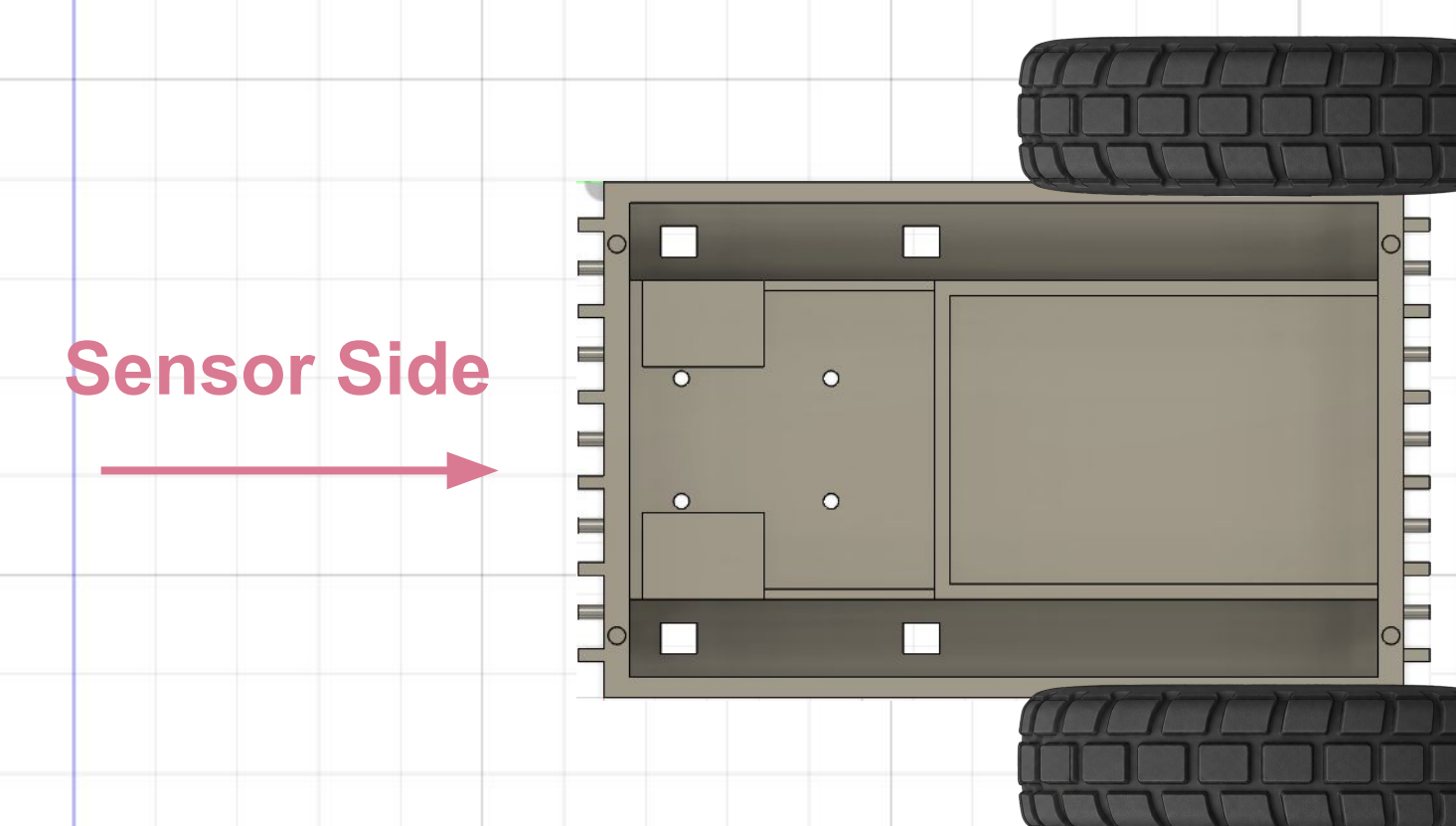
Design Variables Evaluated

Top View: Sensor Placement



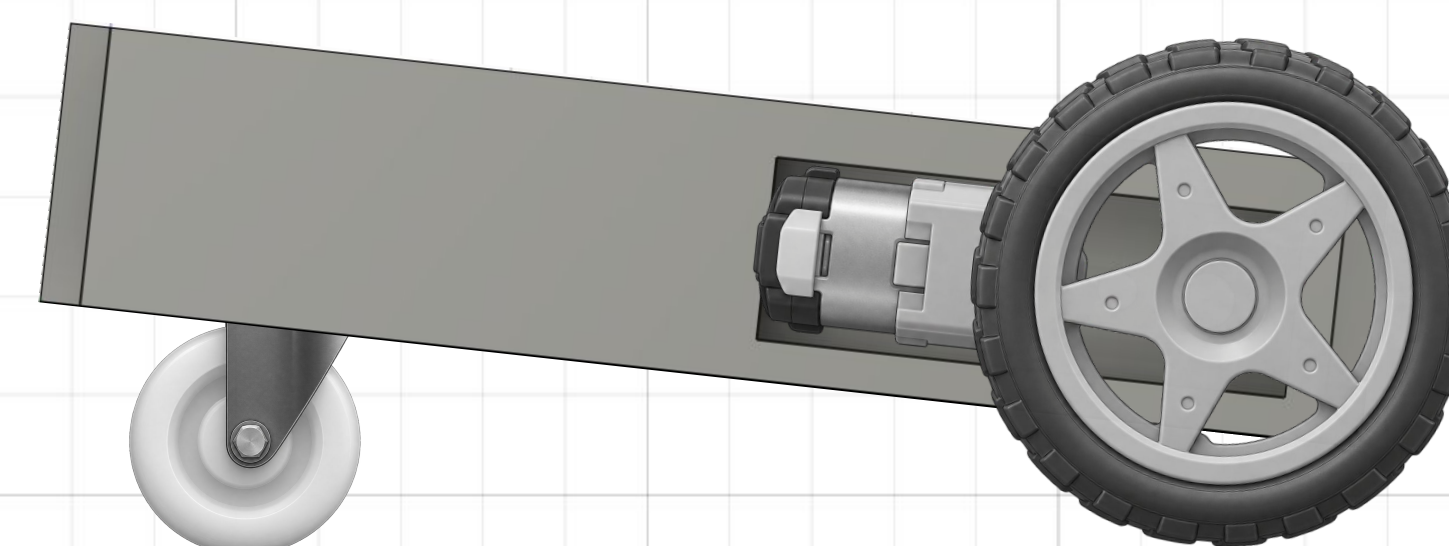
Sensors to drive axle: 16 mm

Top View: Sensor Placement



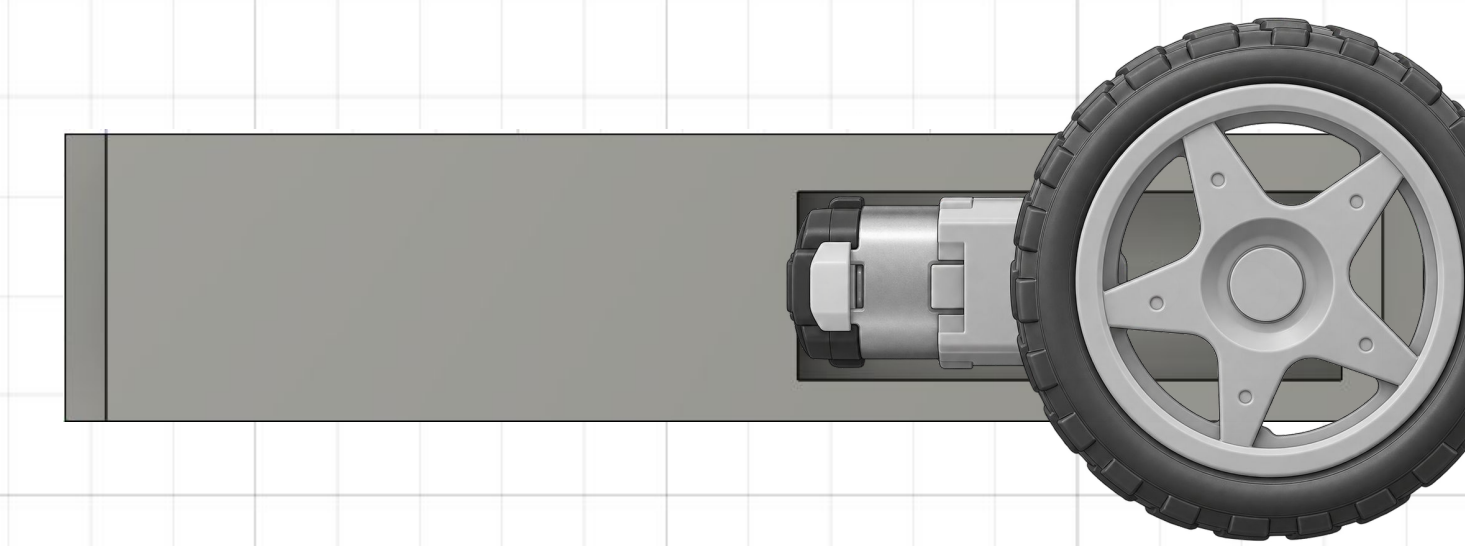
Sensors to drive axle: 139 mm

Side View: Caster Wheel Mounting



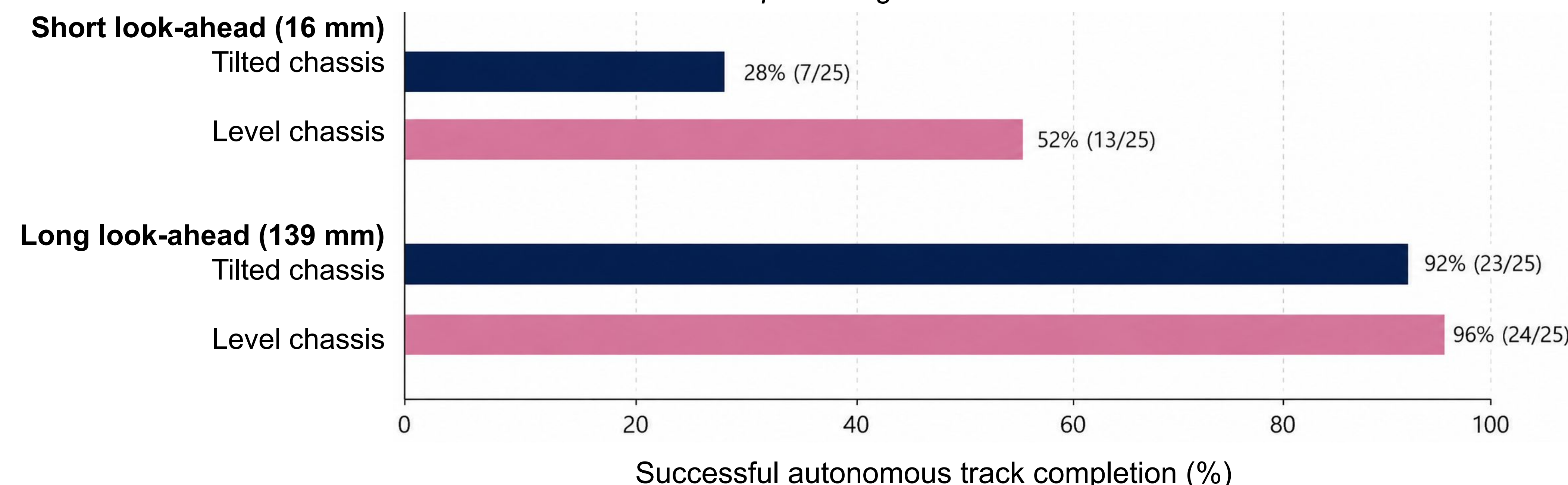
Surface mounted caster; tilted chassis

Side View: Caster Wheel Mounting



Recessed caster; level chassis

Track Completion Performance Across Four Chassis Configurations
n = 25 trials per configuration



Key Findings

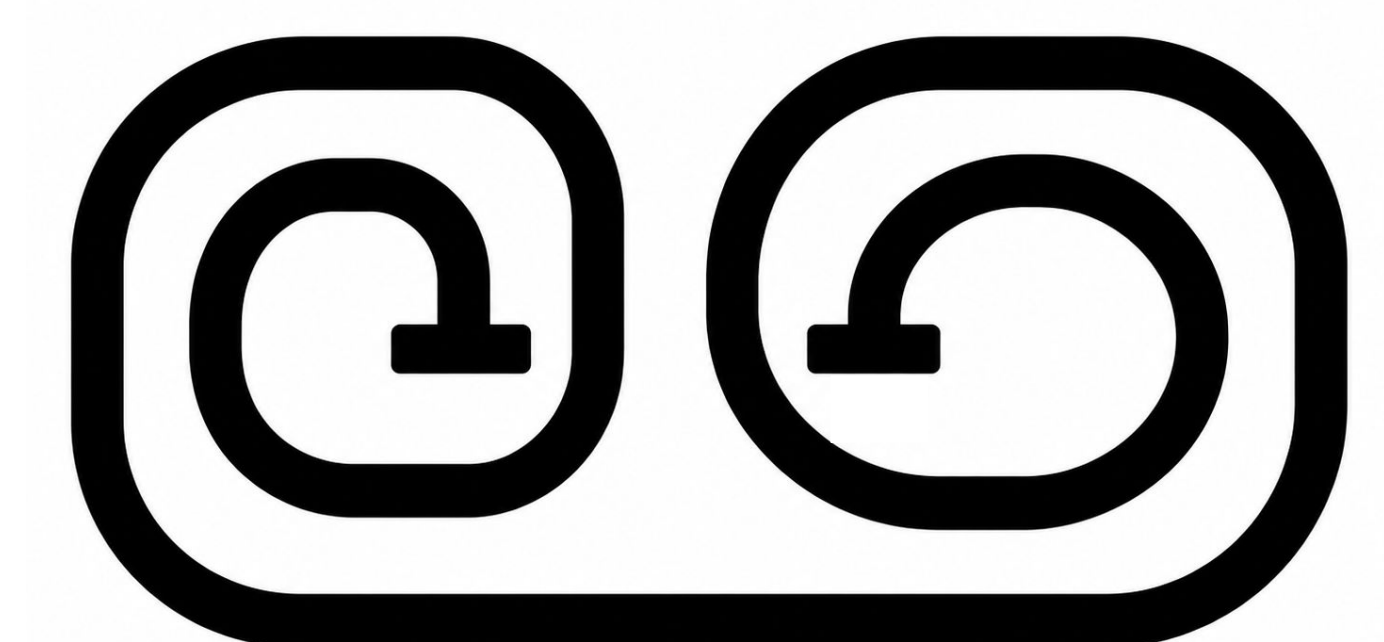
- Increasing the distance from the sensors to the drive axle from 16 mm to 139 mm improved track completion from **28% to 92%** (tilted chassis) and **52% to 96%** (level chassis).
- Level configuration outperformed tilted configuration at both sensor positions
- **Best configuration:** Long look-ahead + level chassis achieved **96% successful autonomous track completion** (24/25 trials).

Conclusion

This project demonstrates that lower-cost drivetrain components can be successfully integrated into an educational autonomous line-following robot. Through both chassis and sensor placement redesigns, the final robot replaces the original four-servo drivetrain with two DC geared motors, a L298N motor driver, and a passive caster wheel to maintain reliable autonomous navigation. Of the four configurations evaluated, the long look-ahead + level-chassis design achieved the highest track completion rate at 96% (24/25 trials). These results support the redesigned robot as a viable lower-cost alternative for next year's summer ECE workshops.

Limitations & Future Work

- Testing was limited to a single track design and can be expanded to more **complex line patterns**.
- **Long term performance** through repeated workshop use have not yet been evaluated.
- Future work can test different motor speeds and supply voltages to **optimize line tracking speed**.



Track design

Acknowledgements

The author would like to thank Professor Hamid Timorabadi, the Department of Electrical and Computer Engineering, and the First Year Summer Research Fellowship for their guidance, support, and access to laboratory resources throughout this project.

References

- [1] F. B. V. Benitti, "Exploring the educational potential of robotics in schools: A systematic review," *Computers & Education*, vol. 58, no. 3, pp. 978–988, 2012, doi: 10.1016/j.compedu.2011.10.006.
- [2] A. Roy and M. M. Noel, "Design of a high-speed line following robot that smoothly follows tight curves," *Computers & Electrical Engineering*, vol. 56, pp. 732–747, 2016, doi: 10.1016/j.compeleceng.2015.06.014.